BLA high-temperature performance — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

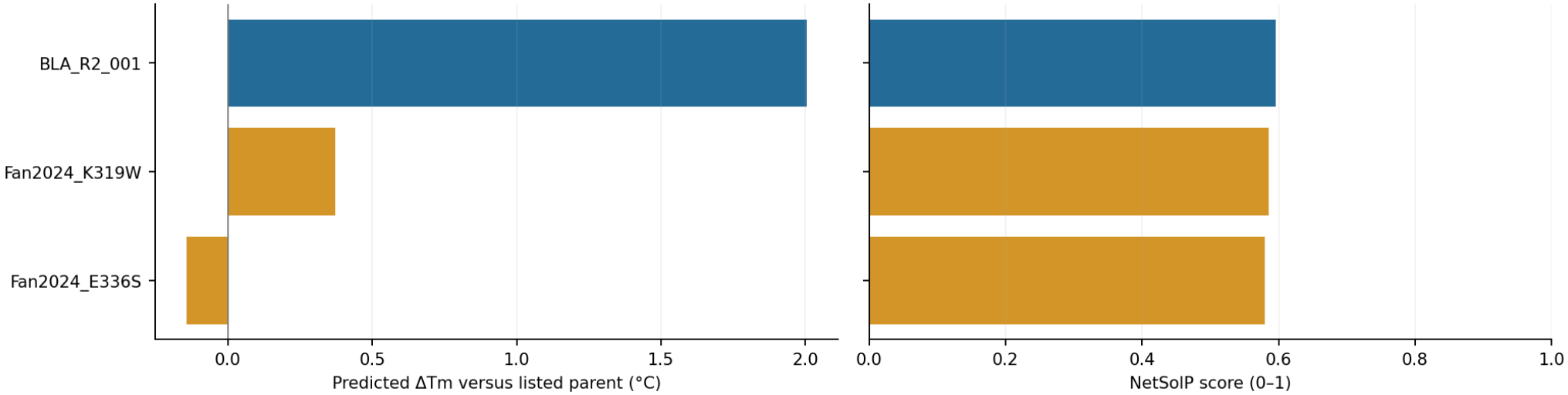
Property / predictor	Published prediction	Our candidates
FoldX stability-change prediction (paper reports kJ/mol)	K319W −1.72 ± 0.08; E336S −1.26 ± 0.11	EvoEF1: BLA_R2_001 +1.48 kcal/mol; matched reference recalculations below.
MD-prepared packing entropy (J K−1 mol−1)	WT 4601; K319W 4614; E336S 4609	Static PACKMAN: BLA_R2_001 7169.33; matched reference recalculations below.

Source: Fan et al., computational screening

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
BLA_R2_001	Our candidate	BLA_PARENT_WP061576039_mature30_512	52.7676	2.0042	0.595761	7.3500
Fan2024_K319W	Reference	BLA_PARENT_WP061576039_mature30_512	51.1367	0.3734	0.585213	7.3357
Fan2024_E336S	Reference	BLA_PARENT_WP061576039_mature30_512	50.6193	-0.1441	0.580433	7.3357

Candidate and reference property predictions



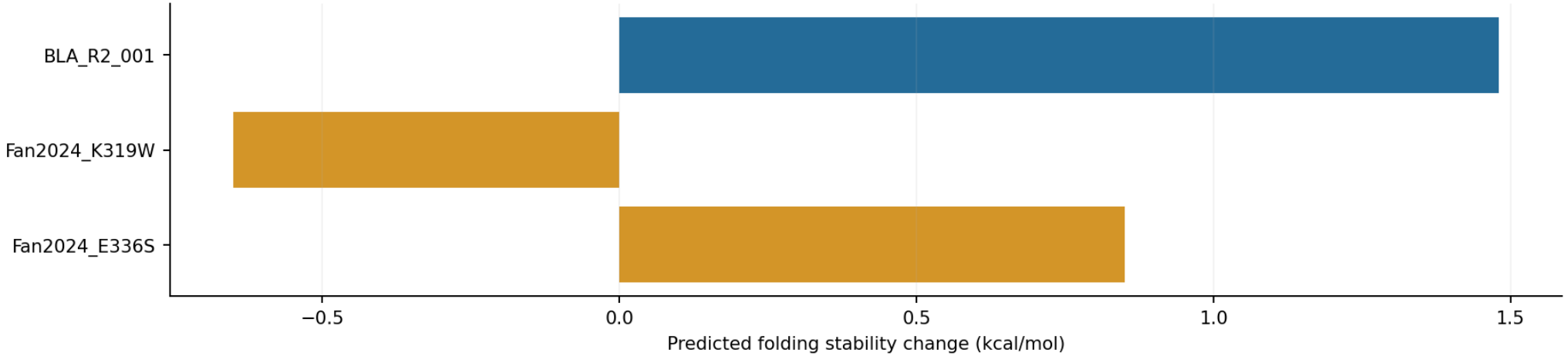
Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Predicted folding stability change — open-source assessment

Method: EvoEF1 1.1. Units: kcal/mol. Protocol: common WT; protein-only; 3 repair runs; 10 matched WT/mutant repacking runs; negative predicts stabilization. Values below were calculated in this work for the listed sequences.

Sequence	Group	Parent / background	Prediction (kcal/mol)
BLA_R2_001	Our candidate	BLA_PARENT_WP061576039_mature30_512	1.48
Fan2024_K319W	Reported reference	BLA_PARENT_WP061576039_mature30_512	-0.65
Fan2024_E336S	Reported reference	BLA_PARENT_WP061576039_mature30_512	0.85

Predicted folding stability change: reference and our sequences



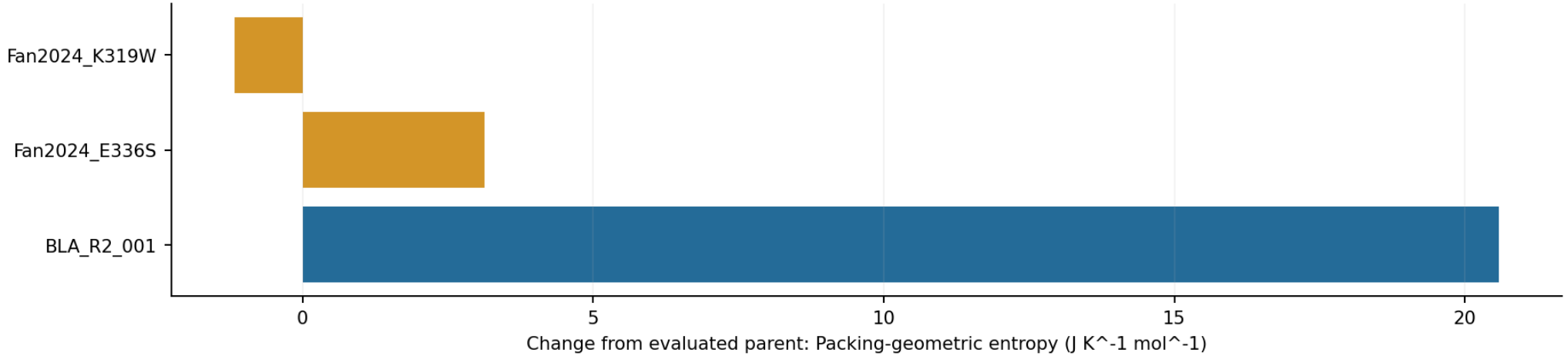
Amber: reported reference variants recalculated here. Blue: our retained candidates. All rows in this panel use the stated method and protocol.

Packing-geometric entropy — open-source assessment

Method: PACKMAN 1.4.13. Units: J K^-1 mol^-1. Protocol: matched fixed backbone; deterministic FASPR all-sidechain packing; 1.4 A probe; 30 sphere points; protein heavy atoms. Values below were calculated in this work for the listed sequences.

Sequence	Group	Parent / background	Prediction (J K^-1 mol^-1)	Change from parent (J K^-1 mol^-1)
Fan2024_K319W	Reported reference	BLA_PARENT_WP06157 6039_mature30_512	7147.58	-1.16801
Fan2024_E336S	Reported reference	BLA_PARENT_WP06157 6039_mature30_512	7151.87	+3.12413
BLA_R2_001	Our candidate	BLA_PARENT_WP06157 6039_mature30_512	7169.33	+20.5858

Packing-geometric entropy: reference and our sequences



Amber: reported reference variants recalculated here. Blue: our retained candidates. All rows in this panel use the stated method and protocol. The plot shows changes from the evaluated parent.

Parent identity, preparation details and run records: results/task_003_bla_high_temperature/agent-alone/README.md.